

GenAI – Hands on 1

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Output Table

Task	Model	Classification (Success/Failure)	Observation (What actually happened?)	Why did this happen? (Architectural Reason)
Generation	BERT	Failure	Model fails to generate normal text; encoder-only model cannot perform causal generation.	BERT is an encoder-only bidirectional model, not trained for next-word generation.
	RoBERTa	Failure	Cannot generate text; no autoregressive head for generation.	RoBERTa is also encoder-only, optimized BERT, but still not a generative model.
	BART	Success	Generated a long continuation, although the text was incoherent and repetitive.	BART is an encoder-decoder model designed for sequence generation.
Fill-Mask	BERT	Success	Predicted "create", "generate", "produce", "develop", etc.	BERT was trained using Masked Language Modeling (MLM) with the [MASK] token.
	RoBERTa	Success	Predicted "generate", "create", "discover", "find", "provide".	RoBERTa uses <mask> token and is heavily optimized for MLM, so predictions are strong.
	BART	Success	Predicted "create", "help", "provide", "enable", "improve".	BART uses <mask> for denoising pretraining, so it can fill masks but less accurately.
QA	BERT	Partial Success / Weak	Returned only "deepfakes," missing other risks.	Base BERT is not fine-tuned for QA tasks like SQuAD, so extraction is weak.
	RoBERTa	Failure	Did not return a meaningful or complete answer.	RoBERTa base model is also not QA-finetuned and cannot reliably extract answers.
	BART	Partial Success	Returned: "Generative AI poses significant risks such as hallucinations," capturing part of the answer.	BART's seq2seq architecture generalizes better but still lacks QA-specific finetuning.

Github Link: <https://github.com/Meghanab1909/GenAI-PES2UG23CS337/tree/main/Hands-On-1>